ARCHIT AGRAWAL

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PROFESSIONAL EXPERIENCE

EdPlus **05/2026 - Present**
AI Software Engineer *Phoenix, AZ*

- Own the architecture, roadmap and delivery of a 12-agent AI analytics platform on AWS Bedrock and Step Functions that turns any structured dataset into cited, verified answers; TypeScript, NestJS, Nuxt.
- Cut wrong answers 85% on 164 published benchmark figures by reading every answer back in plain words and verifying it against the question before display.
- Extended analysis to statistical and causal questions with 25 tests and six causal estimators validated against scipy and statsmodels, scoring 68% on StatQA versus GPT-4o's best reported 64.83%.
- Quadrupled the insights automated reports surface, 13% to 52–59% on unseen datasets, by having an LLM agent choose the analyses while deterministic code computes every number.
- Eliminated a silent failure mislabeling 1 in 8 rows in the prior pipeline, reaching 100% join integrity with schema-constrained LLM output.
- Scaled labeling as a distributed Step Functions fan-out across 64 parallel lanes with a 5% failure circuit breaker, after tracing silent row loss in a run that reported success.

MyStage Music Inc. **07/2025 - 06/2026**
Founding AI/ML Engineer *Remote*

- Re-architected a 14-Cloud-Function backend into one checkpointed LangGraph agent in Python on GCP Cloud Run, cutting production service dependencies 64%.
- Scaled ingestion to 70,000+ records a day from 650+ venues with a Playwright pipeline, transactional locking and Gemini entity resolution that raised search-dataset accuracy 25%.
- Delivered sub-50 ms search on FastAPI and Algolia with real-time Firestore sync, 200+ pytest tests and Logfire tracing.

EdPlus **09/2023 - 05/2025**
Software Engineer *Phoenix, AZ*

- Led a multi-tenant RAG assistant, React to retrieval layer, used by 1,000+ faculty to build courses for 60,000+ students.
- Cut transcript analysis 16x, four hours to 15 minutes, with a Neo4j knowledge graph queried by validated LLM-written Cypher.
- Isolated each college's data at the retrieval layer, so no tenant's material can surface in another's and onboarding became a config change, not a deployment.
- Built a Prompt Flow eval harness scoring groundedness, relevance and coherence to gate every prompt change on results.

Knowledge Exchange for Resilience, Arizona State University **06/2024 - 08/2024**
Data Research Aide *Phoenix, AZ*

- Cut p95 retrieval latency 60% at 95% recall with a hybrid FAISS and Neo4j search layer.
- Built a FastAPI and pgvector recommender reaching 85% top-k precision for collaborator matching, fed by an ETL ingesting 10,000+ profiles a day, plus 20+ REST APIs on .NET 8.
- Lowered API latency 90% with Redis caching and SQL rewrites, and raised dataset accuracy to 95% with PostgreSQL constraints.

Zeus Learning **01/2022 - 07/2023**
Software Engineer *Mumbai, India*

- Built a Redis-backed desk-demand forecasting service used in 300+ Fortune 500 offices, with reported occupancy up 30%.
- Cut deploy time 70% and production incidents 40% by containerizing the CI/CD release pipeline and adding SonarQube gates.
- Migrated a C# .NET monolith to Dockerized Kubernetes microservices, reducing infrastructure cost 20% and footprint 35%.

PROJECTS

SRP Electric MCP Server | GitHub **12/2025 - 01/2026**

- Built a Python MCP server on FastMCP that logs into an undocumented utility portal and gives AI agents 7 tools and 4 analysis prompts for billing, usage, payments and outages, with a one-click setup wizard.

AiJockey – AI DJ Pipeline | GitHub **2025 - Present**

- Built an end-to-end AI DJ in PyTorch and FastAPI: stem separation, an LLM transition planner, 25+ DSP transitions and LUFS mastering, ported off CUDA to an AMD MI300X.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C#
Frameworks: FastAPI, NestJS, Node.js, .NET, React, Vue/Nuxt, Next.js, LangGraph, PyTorch
AI & Data: RAG, LLM evaluation, MCP, AWS Bedrock, PostgreSQL, pgvector, FAISS, Neo4j, Redis, DuckDB, ETL pipelines
Cloud & DevOps: AWS (Step Functions, Lambda, S3, CDK), GCP Cloud Run, Docker, Kubernetes, CI/CD

EDUCATION

Arizona State University (ASU) **08/2023 - 05/2025**
Master of Science, Computer Science *Tempe, AZ, USA*

- **GPA:** 4.0

Dhirubhai Ambani Institute of Information and Communication Technology **08/2018 - 05/2022**
Bachelor of Technology, Information and Communication Technology *India*